



BIOLOGY CH.14 — NERVOUS SYSTEM — FULL MIND MAP

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1. The Brain — Overview & Three Main Parts

- Human brain 3 major parts mein divide hai: Forebrain (complex cognitive activities, sensory-associative functions, voluntary motor activities), Midbrain (body-head movement), aur Hindbrain (breathing, blood pressure, sneezing, swallowing). *[Q785]*
- 'Fore-brain' brain ka main thinking part hai — voluntary actions, thinking aur information processing karta hai. Midbrain eye-movement, auditory-visual processing karta hai. Hindbrain breathing-heart-blood vessels control karta hai. *[Q811, Q816]*
- Brain body ka 'main' coordination center hai — voluntary movements, body balance, involuntary organs (lungs, heart, kidneys), thermoregulation, hunger-thirst, circadian rhythms control karta hai. *[Q808]*

2. Spinal Cord & Reflex Actions

- 'Spinal cord' nerve fibres se bana hota hai jo reflex actions mediate karte hain aur brain se impulses transmit karte hain — Central Nervous System ka part hai, long-thin-tubular structure. *[Q787]*
- 'Aag se haath khinch lena' — ye reflex action 'Spinal Cord' se control hota hai — Cerebrum conscious thought ke liye, Cerebellum balance-posture ke liye hota hai. *[Q810]*

3. Neuron — Structure & Components

- Neuron mein information 'Dendrite' mein acquire hoti hai — Dendrite electrical impulses ko cell body ki taraf conduct karta hai, Axon impulses ko cell body se door le jaata hai. *[Q788]*
- ⚠ Trap: 'Cone' neuron ka part NAHI hai — ye EYE ka photoreceptor cell hai. Neuron ke 3 main parts: Dendrites, Axon, Cell body (Soma). *[Q791]*
- Neuron ka cell body 'Soma' (ya Perikaryon) kehlaata hai — macromolecules synthesis ka main site hai. Schwann cells myelin produce karti hain, Dendrons soma ki branch-like extensions hain. *[Q793]*

- 'Dendrons' Neuron ke parts hain — dendrites nerve impulses ko cell body ki taraf carry karti hain, Axon impulses ko cell body se synapse tak carry karta hai. Nephron kidney ki unit hai. *[Q796]*

4. The Synapse

- 'Synapse' wo point hai jahan nervous impulse ek 'Neuron se doosre Neuron' tak pass hoti hai — Neurons nervous system ki basic units hain, types: Sensory, Motor, Interneurons. *[Q789]*
- 2 neurons ke beech ki junction 'Synapse' kehlaati hai — presynaptic-postsynaptic neuron ki membranes se banti hai, electrical-chemical types hoti hain. *[Q795]*
- Axon ending aur doosre nerve cell ke dendrite ke beech ki junction 'Synapse' kehlaati hai. *[Q800]*
- 2 neurons ke beech ki space 'Synapse' kehlaati hai — Dendrite/Axon neuron ke hi parts hain, CSF plasma ka ultrafiltrate hai. *[Q812]*

5. Hindbrain — Pons, Medulla & Involuntary Control

- Vomiting jaisi involuntary action 'Hindbrain' (Rhombencephalon) control karta hai — medulla oblongata, pons aur cerebellum se bana hota hai, respiratory rhythm-motor activity-sleep control karta hai. *[Q786]*
- 'Pons' human brain ka part hai — Hindbrain mein Pons, Cerebellum, Medulla shaamil hain. Pons fibre tracts se bana hai jo brain regions ko interconnect karte hain. *[Q790]*
- Involuntary actions (blood pressure, salivation, vomiting) 'Medulla' se control hoti hain — respiration, cardiac function, vasodilation, reflexes (swallowing-sneezing) bhi. *[Q813, Q818]*
- Breathing 'Medulla' se regulate hoti hai — brainstem aur spinal cord ke beech connection hai, autonomic functions (breathing, heartbeat, digestion) regulate karta hai. *[Q820]*

6. EEG (Electroencephalogram) Test

- 'EEG' test 'Brain' ki ailments diagnose karne ke liye conduct hoti hai — scalp par electrodes lagakar epilepsy, sleep disorders, brain tumours diagnose karti hai. Hans Berger ne discover ki. *[Q792]*

7. The Auditory Nerve

- Human ear sound ki audible frequencies ko electric signals mein convert karke 'Auditory Nerve' se brain tak bhejta hai — cochlea (inner ear) se directly brain tak jaata hai. *[Q794]*

8. Corpus Callosum

- Mammals mein 'Corpus Callosum' 2 Cerebral Hemispheres ko connect karta hai — depth perceive karne mein help karta hai, 4 parts mein divide hai: rostrum, genu, body, splenium. *[Q797]*

9. Brain Weight & Physiology

- Brain ka weight ~1.4 kg hota hai, aur har minute 0.85 litres blood isse guzarta hai — 3 major parts: forebrain (cerebrum, thalamus, hypothalamus), midbrain, hindbrain. [Q798]
- Human brain ka weight 1200-1400 grams hota hai — Adult male ~1370 grams, Adult female ~1200 grams. [Q809]

10. Nerve Cell (Neuron) Length

- Ek individual nerve cell 100 cm (1 meter) tak lambi ho sakti hai — neurons human body ki longest cells hain, longest axons sciatic nerves ke hote hain. [Q799]

11. The Cerebellum

- Cerebellum 'coordination of muscular movements' se related hai — Cerebrum conscious-unconscious mind, intelligence, memory control karta hai; Medulla oblongata involuntary actions (heartbeat, breathing) control karta hai. [Q801]
- ⚠ Trap: Cerebellum ke 2 parts 'Cerebral hemispheres' NAHI kehlaate — inhe 'Cerebellar hemispheres' kehte hain. Cerebellum posterior region mein hai, hindbrain ka part hai, voluntary actions ki precision-posture-balance ke liye responsible. [Q806]
- ⚠ Trap: Cerebellum brain ka '25%' NAHI, sirf ~10-12% hota hai — voluntary actions ki precision aur posture-balance maintain karne ke liye responsible hai, hindbrain ka part hai. [Q807]
- Cerebellum ('little brain') voluntary actions ki precision aur body ki posture-balance maintain karne ke liye responsible hai — spinal cord aur brain ke doosre parts se information receive karta hai. [Q819]

12. The Cerebrum

- ⚠ Trap: Cerebrum 'two lateral cerebellar convolutions aur central vermix' se NAHI bana — ye description actually CEREBELLUM ki hai. Cerebrum forebrain ka part hai, brain ka largest part (~80%) hai. [Q802]
- ⚠ Trap: Cerebrum brain ka '60%' NAHI, ~80% constitute karta hai — forebrain ka part hai, 2 cerebral hemispheres hote hain, brain ka front-superior-lateral side banata hai. [Q803]
- ⚠ Trap: 'Cerebrum muscle activity coordinate karta hai' — ye statement INCORRECT hai. Muscle activity coordination primarily CEREBELLUM ki responsibility hai (brain ke peeche located), Cerebrum thinking-memory-perception mein crucial role play karta hai. [Q805]

13. Voluntary vs Involuntary Muscles

- Voluntary muscles (skeletal muscles) consciously control ho sakti hain (jaise frog ka jump karna, handwriting) — Involuntary muscles bina soch work karti hain (stomach mein food movement, heartbeat). [Q804]
- ⚠ Trap: 'Cycling' involuntary action NAHI hai — conscious choice aur muscle control chahiye. Salivation, respiration, digestion involuntary actions hain (medulla oblongata se control hoti hain). [Q814]
- ⚠ Trap: 'Heart ka beating' voluntary action NAHI hai — spinal cord se control hoti hai. Fist band karna, chair move karna, object uthaana voluntary actions hain (will/brain ke control mein). [Q817]

14. Nodes of Ranvier

- 'Nodes of Ranvier' myelinated axons mein microscopic gaps hote hain — Schwann cells ke beech chhote gaps hote hain, nerve impulse ki saltation allow karte hain. [Q815]